



HAMMARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: ENS 221
Mid-Term Examination
Date: 12/5/2025

Course Title: Environmental Science
Semester: Fall 2025
Total Marks: 30

Total Time: 1:30 Hours

* If necessary, draw relevant graphs, diagrams to support your explanations.: $3 \times 10 = 30$
[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

CLOs Marks

Part A (Answer any One)

- Q1. (a) What do you mean by Environment? CLO1 [2]
(b) Differentiate between environmentalist and scientists with examples. CLO2 [4]
(c) Discuss the significance of both fields in addressing global environmental challenges. CLO3 [4]
- Q2. (a) What is terrestrial and aquatic ecosystem with its example? CLO1 [3]
(b) "Environmental Science is a multidisciplinary approach" Explain it. CLO2 [4]
(c) Describe the largest ecosystem of the earth and what is the main function of ecosystem? CLO3 [3]

Part B (Answer any Two)

- Q3. (a) What is the water cycle? CLO1 [1]
(b) What is the role of wetlands in maintaining the water cycle? CLO2 [2]
(c) Briefly discuss the process of hydrological cycle with its importance. CLO3 [4]
(d) Explain the First Law of Thermodynamics with equation. CLO2 [3]
- Q4. (a) Define the greenhouse effect? CLO1 [1]
(b) Identify sources of greenhouse gases and discuss their impact on the environment. CLO2 [4]
(c) Draw and explain the layers of the atmosphere, with its importance. How do they each function in Earth's atmosphere? CLO3 [5]
- Q5. (a) Define food security? CLO1 [1]
(b) List the five main indicators of food security and explain how climate change can affect food security. CLO2 [3]
(c) What is sustainable agriculture? Describe its key principles and why it is important for both food security and the environment. CLO3 [6]



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE225: Digital Logic Design

Batch: 26th

Date: 11/12/2025

Midterm Examination

Semester: Fall 2025

Time: 1 Hour 30 minutes

Total Marks: 30

Answer any five questions of the following.

5 × 6 = 30

1. (a) Find the 9's and 10's complements of the number $(98376)_{10}$. CLO3 2
(b) Perform the following operations using complements showing all steps clearly: CLO3 4
 - i. $(1422)_5 - (3132)_5$ using 4's complement
 - ii. $(110101)_2 - (101011)_2$ using 2's complement
2. (a) Distinguish between Minterms and Maxterms. CLO3 2
(b) Simplify the following expression to the minimum numbers of literals. CLO3 2
$$F = (W+X)(W+X')(Y+Z)(Y'+Z)$$

(c) Prove using Boolean algebra: $(X+Y)'Z + XY' = Y'(X+Z)$ CLO2 2
3. (a) Derive POS from the following simplified SOP using truth table. CLO2 3
$$F = A + B'C + A'B$$

(b) Reduce the following function to exactly 6 literals. CLO3 3
$$F(W, X, Y, Z) = W'X'Y + WXY' + W'XZ + WX'Z'$$
4. (a) Simplify the following function using k-map: CLO3 3
$$F = \sum m(2, 4, 6, 7, 9, 10) + d(8, 11, 12, 13)$$

(b) For the following function find out the Sum of Minterms (SOM): CLO2 3
$$F = X' + X(X+Y')(Y+Z')$$
5. Given the following function: CLO2 6
$$F = \prod M(6, 7, 8, 9, 13, 14, 15)$$
 - i. Find out the POS expression using truth table CLO5
 - ii. Find the reduced POS
 - iii. Draw the logic diagram of the reduced POS and calculate the cost of the circuit.

6/20
ABD'

6. Consider the following function:

$$F = \sum m(1, 5, 6, 8)$$

- i. Create the truth table and expression in SOP
- ii. Find the simplified SOP
- iii. Draw the logic diagram of the simplified SOP and calculate the cost of the circuit.

CLO1 6
CLO5

7. For the function $F(A, B, C, D) = \sum m(0, 2, 5, 6, 7, 8, 10, 13, 14, 15)$

- i. Identify ALL prime implicants
- ii. Identify ALL essential prime implicants
- iii. If the above function had don't care conditions at $m(1, 4, 12)$, how would the prime implicants and essential prime implicants change?

CLO3 6



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology Department of Computer Science and Engineering

CSE 227: Data Communication

Batch: 26th
Date: 14/12/2025

Mid-Term Examination

Semester: Fall 2025
Time: 1 Hour 30 Minutes
Total Marks : 30

Answer all the questions:

5 × 6 = 30

	CLOs	Marks
✓ 1. (a) Explain the five components of a data communication system.	CLO1	3
(b) Write the three criteria necessary for an effective and efficient network.	CLO1	3
✓ 2. (a) Write the differences between the duties of the IETF and IRTF.	CLO4	2
(b) For each of the following four networks, discuss the consequences if a connection fails.	CLO1	4
i. Five devices arranged in a mesh topology		
ii. Five devices arranged in a star topology (not counting the hub)		
iii. Five devices arranged in a bus topology		
iv. Five devices arranged in a ring topology.		
✓ 3. (a) Write the reasons for OSI Model's lack of success.	CLO4	3
(b) Explain Multiplexing and De-multiplexing in TCP/IP protocol suite.	CLO5	3
✓ 4. (a) List some protocols used in application-layer and transport-layer.	CLO4	3
(b) Describe Nyquist Bit Rate and Shannon Capacity.	CLO3	3
5. (a) Distinguish between baseband transmission and broadband transmission.	CLO3	3
(b) Explain Amplitude, Frequency and Phase.	CLO3	3
✓ 6. (a) Name and explain three types of transmission impairment.	CLO3	3
(b) A device is sending out data at the rate of 1000 bps.	CLO2	3
i. How long does it take to send out 10 bits?		
ii. How long does it take to send out a single character (8 bits)?		
iii. How long does it take to send a file of 100,000 characters?		
7. (a) Briefly explain five line coding schemes.	CLO2	3
(b) Explain Baseline Wandering and DC Components.	CLO2	3



HAMMARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering

CSE221: Computer Algorithms

Mid-Term Examination

Semester: Fall 2025

Time: 1 Hour 30 Minutes

Total Marks: 30

Batch: 26th

Date: 07-12-2025

Answer any five questions of the followings.

5 × 6 = 30

CLOs Marks

1. (a) What is algorithm analysis and why is it important? CLO1 2
(b) Analyze the complexity of the following recurrence: CLO1 4

$$T(n) = \begin{cases} 2T(n-1) - 1 & \text{if } n > 1 \\ 1 & \text{otherwise} \end{cases}$$
2. (a) What is the worst-case time complexity of Quick Sort and when can this happen? CLO2 2
(b) Apply Quick Sort Algorithm and find the appropriate index for item 36. CLO2 4
 $A[9]: 36, 25, 37, 23, 42, 14, 47, 53, 35$
3. (a) Define feasible solution and optimal solution. CLO1 2
(b) Suppose you have 10 elements in an array as follows: CLO2 4

Address	1	2	3	4	5	6	7	8	9	10
Elements	27	-8	35	23	85	-17	25	31	2	12

Draw the recursion tree to find the maximum and minimum elements from the array using recursive Min-Max Algorithm.
4. (a) How does divide and conquer improve performance over brute force? CLO2 1
(b) Find the contiguous maximum subarray of the following array using Divide & Conquer approach: CLO2 5
 $\{4, -5, 2, 4, -1, 3\}$
5. (a) Explain the components of Greedy Method. CLO3 2
(b) You are given the following 8 intervals: CLO5 4
 $A: (2, 4), B: (3, 5), C: (4, 9), D: (5, 8), E: (6, 7), F: (8, 11), G: (9, 13), H: (10, 12)$
Schedule the intervals using Greedy Method.
6. (a) What is the role of sorting in greedy algorithms? Why is it often necessary? CLO3 2
(b) Consider the following instance of knapsack problem: CLO5 4
Number of objects, $n = 6$;
Capacity of knapsack, $m = 17$;
Profits, $\{p_1, p_2, p_3, p_4, p_5, p_6\} = \{10, 5, 18, 17, 21, 18\}$ and
Weights, $\{w_1, w_2, w_3, w_4, w_5, w_6\} = \{3, 2, 4, 2, 4, 5\}$.
Calculate the optimal solution using Greedy Method.

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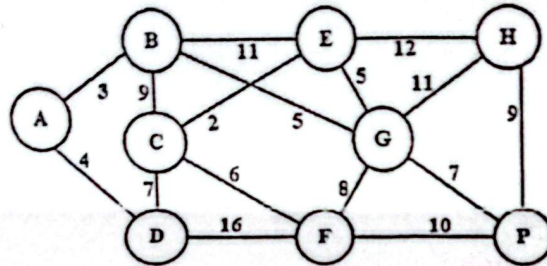
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2



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7. (a) When is Prim's algorithm preferred over Kruskal's algorithm? CLO3 2
- (b) Apply Kruskal's Algorithm to generate the minimum spanning tree for the following graph: CLO5 4





HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 223: Object Oriented Programming

Batch: 26th

Semester: Fall 2025

Time: 1 Hour 30 Minutes

Date: 09/12/2025

Mid-Term Examination

Total Marks: 30

Answer five questions of the following.

5 × 6 = 30

	CLOs	Marks
1(a). Identify the three literals that define value in Java, and briefly explain two rules concerning the starting characters and case usage for naming a valid Java identifier with examples.	3	2
1(b). Write the primary role of the Java Virtual Machine (JVM) in the compilation process, and list the core steps involved in the JVM working procedure.	3	2
1(c). Write the differences between the Java Development Kit (JDK) and the Java Runtime Environment (JRE).	3	2
2(a). Why is the size of char data type 2 bytes in Java?	1	2
2(b). Briefly explain type casting with the example.	1	2
2(c). Consider the following program:	2	2

```
class Main {  
    public static void main(String[] args) {  
        boolean a = ? ;  
        boolean b = ? ;  
        System.out.println(a || b);  
        System.out.println(a && b);  
    }  
}
```

Write every possible output for the operators a and b.

3(a). What happens if a program attempts to use an illegal index of an array?	5	2
3(b). The Stanton measure of an array is computed as follows: i. count the number of 1s in the array ii. let this count be n iii. measure the number of times that n appears in the array	2	4

For example, the Stanton measure of {1,4,3,2,1,2,3,2} is 3 because 1 occurs 2 times in the array and 2 occurs 3 times. Write a method in Java named `stantonMeasure()` that returns the Stanton measure of its array argument.

- 4(a). Explain access modifiers and related member variable scopes in Java. 3 2
- 4(b). Consider the following code: 2 2

```

{
    int x = 5;
    {
        int x = 10;
        System.out.println(x);
    }
}

```

Write the output of the given program. If an error occurs, correct the code and then write its output.

- 4(c). Briefly explain 'Local Variables' using 'this' keyword. 3 2

- 5(a). Briefly explain the **nested try block** in exception handling. 5 2

- 5(b). Write a Java program that asks the user for a number and displays the number multiplied by 10. Also the program must throw an exception when the user enters a value greater than 10. 5 4

- 6(a). Explain Constructor Overloading with an example program? 1 3

- 6(b). Write a function named **nextPerfectSquare** that returns the first perfect square that is greater than its integer argument and less than 100. A **perfect square** is an integer that is equal to some integer squared. For example, 16 is a perfect square because $16 = 4 \times 4$. However 15 is not a perfect square because there is no integer n such that $15 = n \times n$. 2 3

- 7(a). What are the key differences between **ArrayList** and **Array**? 1 2

- 7(b). Consider an **ArrayList** [A,B,C,D,E]. Write a Java program to insert an element into the array list at the first position and remove the fifth element from the list. 3 4



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering

CSE229: Computer Architecture

Batch: 26th

Date: 17/12/2025

Midterm Examination

Semester: Fall 2025

Time: 1 Hour 30 minutes

Total Marks: 30

Answer any five questions of the following.

5 × 6 = 30

- ✓ (a) Distinguish between Computer Architecture and Computer Organization with examples. CLO1 3
- (b) Draw the hierarchical structure of a computer system showing Structure and Function at each level. CLO1 3

2. (a) State the overflow rule with one example each. CLO4 2

- (b)

```
void strcpy (char x[], char y[]) {  
    int i; i = 0;  
    while ((x[i] = y[i]) != '\0')  
        i += 1;  
}
```

 CLO4 4

What is the compiled MIPS assembly code?

3. (a) $A[3000] = h + A[3000]$; Write down MIPS Instruction and make MIPS code table from the instruction. CLO4 3
- (b) Explain Booth's algorithm principle and its advantages over standard multiplication. CLO5 3

- X (a) Simulate $(-5) \times (3)$ using Booth's algorithm with 4-bit representation. Show complete register traces. CLO4 3
- (b) In the following code segment, f, g, h, i, and j are variables. If the five variables f through j correspond to the five registers \$s0 through \$s4, what is the compiled MIPS code for this C if statement? And also draw the flowchart. CLO4 3

```
if (i == j) {  
    f = g + h;  
}  
else {  
    f = g - h;  
}
```

5. (a) Draw a flowcharts for Floating-point division. CLO3 2
(b) Draw a flowchart for floating-point addition/subtraction showing the normalization steps. CLO3 4

6. (a) Draw the complete instruction cycle with interrupts state diagram showing all possible states and transitions. CLO3 2

(b) while (save[i] == k) CLO3 4

i += 1;

Assume that i and k correspond to registers \$S3 and \$S5, and the base of the array save is in \$S6. What is the MIPS assembly code corresponding to this C segment?

7. (a) int leaf_example (int g, int h, int i, int j) { CLO3 3
 int f;
 f = (g + h) - (i + j);
 return f;
}

- (b) Describe the underlying principles of hardware design CLO3 3



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE229: Computer Architecture

Batch: 26th
Date: 10/03/26

Final Examination

Time: 2 Hour
Total Marks: 40

Answer any **five** questions of the following

No	Question/ Statement	CLOs	Marks
1.	(a) Distinguish between Computer Architecture and Computer Organization with examples.	CLO1	2
	(b) Draw the internal organization of a 2M * 8 dynamic memory chip.	CLO1	6
2.	(a) Define RAM. Write down the types of RAM.	CLO4	2
	(b) Draw the diagram of synchronous DRAM	CLO4	6
3.	(a) Write about EEPROM.	CLO4	2
	(b) Draw the diagram of the Error Detection/Correcting Mechanism.	CLO5	6
4.	(a) Draw the Pentium 4 block diagram.	CLO4	4
	(b) Write down three replacement algorithms.	CLO4	4
5.	(a) What is a hazard?	CLO3	2
	(b) Write down the types of hazards. To overcome these hazard which steps would you take?	CLO3	6
6.	(a) Draw the flowchart of a Six Stage Instruction Pipeline.	CLO3	4
	(b) Define Instruction pipeline. Write down the steps of the Instruction pipeline.	CLO3	4
7.	(a) Write about the Membrane Key switch and Hall effect Key switch with proper diagram.	CLO3	3
	(b) Write down the differences between impact printers and non-impact printers. Write down the advantages and disadvantages of dot matrix printers.	CLO3	5



HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: ENS 221
Final Examination
Date: 2/27/2026

Course Title: Environmental Science
Semester: Fall 2025
Total Marks: 40

Total Time: 02:00 Hours

* If necessary, draw relevant graphs, diagrams to support your explanations.: $4 \times 10 = 40$
[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

CLOs Marks

Part A (Answer any Two)

- Q1. (a) Define water pollution? CLO1 [1]
(b) Explain the concept of environmental pollutants and their sources. CLO2 [4]
(c) What measures can be taken to reduce water pollution in Bangladesh, considering the challenges of industrial waste and agricultural runoff? CLO3 [4]
- Q2. (a) Define green tourism? CLO1 [1]
(b) Discuss the impact of ecotourism on local culture and biodiversity. CLO3 [4]
(c) What are the key barriers to sustainable ecotourism in Bangladesh, and how can the country overcome these challenges through policy and community engagement? CLO4 [5]
- Q3. (a) Define SDGs? CLO1 [1]
(b) What steps should Bangladesh take to combat climate change, and how can it contribute to global climate efforts? CLO2 [4]
(c) How does the SDG Pyramid illustrate the interlinkages between sustainability dimensions (spiritual, ecological, people)? CLO3 [5]

Part B (Answer any Two)

- Q4. (a) Define mitigations in disaster? CLO1 [2]
(b) All disasters are hazards, but all hazards are not disasters. CLO2 [3]
(c) Discuss the effectiveness of cyclone preparedness programs (CPP) in Bangladesh. CLO3 [5]
- Q5. (a) What is SLR? CLO1 [1]
(b) What are the main causes of climate change in Bangladesh? Provide it. CLO2 [3]
(c) What are the impacts of climate change on water resources and floods in Asia, and how can Bangladesh prepare for these? CLO2 [6]

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- Q6. (a) What is the role of early warning systems in disaster management? CLO1 [2]
- 34 (b) Explain the importance of disaster preparedness, response, and recovery. CLO3 [3]
- (c) Describe the First Three Pillars of Bangladesh's Climate Change Strategy and Action Plan. How do they help improve disaster management and strengthen local institutions? CLO4 [5]



HAMMARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE225: Digital Logic Design

Batch: 26th
Date: 01/03/2026

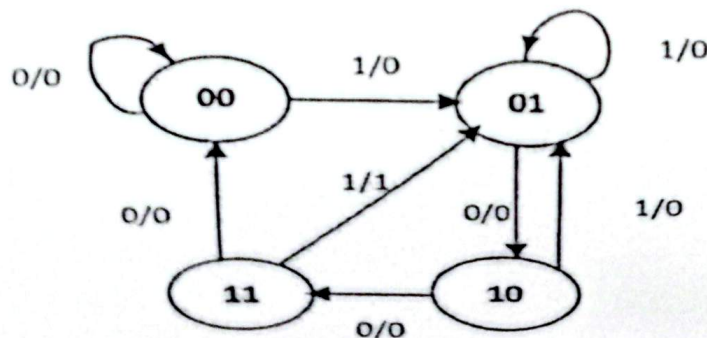
Final Examination

Semester: Fall 2025
Time: 2 Hours
Total Marks: 40

Answer any five questions of the following.

5 × 8 = 40

1. (a) Graphically explain Sequential Circuit in terms of memory elements and time dependency. CLO1 2
- (b) Write the limitations of latches and justify why modern digital systems prefer flip-flops. CLO1 2
- (c) Draw and explain the implementation of an SR latch with NAND and NOR configurations. Validate its operation for all input combinations using truth tables. CLO1 4
2. (a) When does Race-Around Condition appears in a JK flip-flop? How can it be resolved? CLO1 3
- (b) Convert the SR latch into SR flip-flop and derive its characteristics table and excitation table using truth table. CLO1 5
3. (a) Derive the necessary logic configuration to transform a SR flip-flop into a T flip-flop. CLO5 4
- (b) Demonstrate how a JK flip-flop can be converted into a SR flip-flop. CLO5 4
4. (a) Derive the equations using circuit excitation table for the given state diagram that are necessary to design a clocked sequential circuit using a T flip-flop. CLO5 4



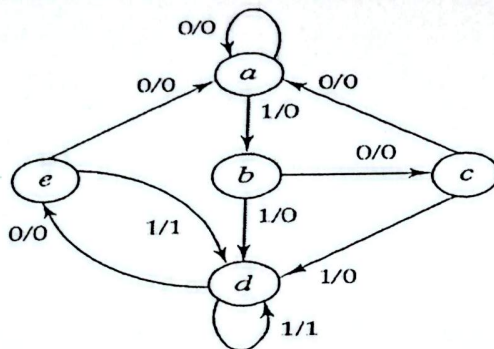
- (b) Perform state assignment and state reduction on the following state table. Draw the reduced state diagram. CLO5 4

State	Next State		Output	
	x=0	x=1	x=0	x=1
001	000	001	0	0
010	010	011	0	0
011	000	011	0	0
100	100	011	0	1
101	000	011	0	1
101	000	011	0	1

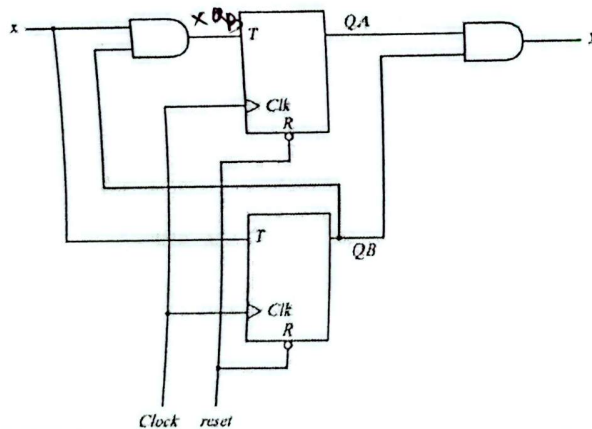
5 (a)

Derive the state table for the following state diagram.

CLO3 3



- (b) For the given sequential circuit, analyze the possible state transitions, output sequences and finally draw a state diagram. CLO5 5



6. (a) Design a 4-bit asynchronous DOWN counter using JK flip-flop. CLO5 3
 (b) Design a 3-bit asynchronous UP/DOWN counter and provide a logical explanation of its operation. CLO5 5
 7 (a) What are Programmable Logic Array and Programmable Array Logic? CLO4 2
 (b) Given the Boolean functions $Y_1=AB$ and $Y_2=AC+BC'$, design a PAL that implements these functions. CLO5 3
 (c) Design the block diagram of an 32x8 ROM and explain how it works. CLO5 3



HAMMARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology Department of Computer Science and Engineering CSE 223: Object Oriented Programming

Batch: 26th

Date: 03/03/2026

Final Examination

Semester: Fall 2025

Time: 2 Hours

Total Marks: 40

Answer any five questions of the following.

5 × 8 = 40

	CLOs	Marks
8/1(a). What are autoboxing and unboxing? Give an example for each of them.	3	4
1(b). Write the differences between autoboxing and unboxing.	3	4
2(a). Briefly explain Java Abstraction with example.	1	3
2(b). Write a Java program to create an abstract class Animal with an abstract method called sound() . Create subclasses Lion and Tiger that extend the Animal class and implement the sound() method to make a specific sound for each animal.	2	5
3(a). What is operator overloading in object-oriented programming?	1	2
3(b). Define polymorphic variables. Explain polymorphic variables with example.	1	6
6/4(a). Explain the life cycle of a Thread in object-oriented programming.	3	4
4(b). Write two different ways to create Threads in Java. Which one is better and why?	3	4
5/5(a). 'Java strings are immutable.' - justify this statement.	3	4
5(b). Explain the needs for interface in Java.	3	4
4/6(a). Why is super() method used in object-oriented programming?	5	2
6(b). Explain the types of Inheritance in Java.	1	6
7/7(a). Define Abstract Window Toolkit.	4	2
7(b). Write a JavaAWT program to create a simple calculator for addition and division operation between two integer numbers.	4	6



HAMMARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE221: Computer Algorithms

Batch: 26th

Date: 05-03-2026

Final Examination

Semester: Fall 2025

Time: 2 Hours

Total Marks: 40

5 × 8 = 40

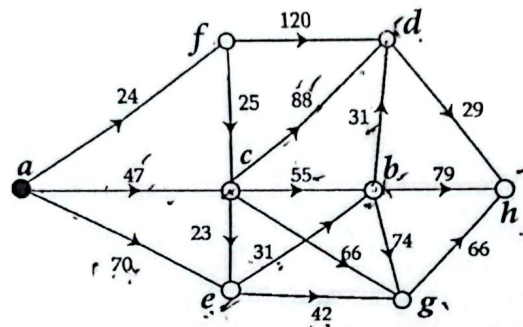
CLOs Marks

CLO4 3

CLO3 5

Answer any five questions of the followings.

1. (a) Write the differences between Greedy Method and Dynamic Programming.
- (b) Apply Dijkstra's Algorithm to find the shortest path from the node 'a' to all other nodes in the graph.

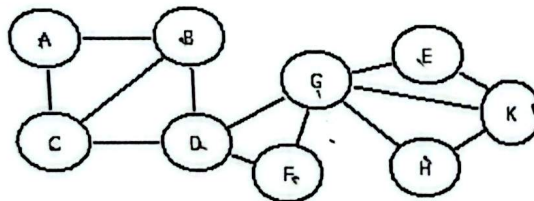


2. (a) A social media platform wants to find all connected users starting from one user. Explain how a Breadth-First Search (BFS) can help.

CLO2 2

- (b) Find the articulation points of the following graph using a Depth-First Search (DFS) spanning tree.

CLO3 6



3. (a) Suppose, a warehouse manager wants to maximize profit by selecting products to load into a truck with a limited weight capacity. Each product has a weight and profit value. He must decide which products to include. Which Dynamic Programming approach should be used, and why?

CLO2 3

- (b) Assume, a system has three devices with the following costs and reliabilities:

CLO3 5

$$\{c_1, c_2, c_3\} = \{30, 15, 20\}$$

$$\{r_1, r_2, r_3\} = \{0.9, 0.8, 0.5\}$$

Design a three-stage system with a maximum cost of 105 and show the reliability.

4. (a) A software engineer is developing a program to compute the nth Fibonacci number. The recursive version becomes very slow for a large value of n. Compare Recursion and Dynamic Programming for this scenario.

CLO4 3

- (b) Suppose, you have two strings A = "BQDRCVEFGH" and B = "ABCVDEFGH". Now calculate the longest common subsequence (LCS) using Dynamic Programming.

CLO3 5



- 3/5
- (a) A company wants to schedule maximum non-overlapping activities from a list of time intervals. Compare Greedy Method and Backtracking for solving this problem. CLO4 3
- (b) Find three solutions for the **n-queens** problem using backtracking when $n = 5$. CLO3 5
6. (a) OneSolutionBD is selecting projects within a limited budget to maximize total benefit. Explain how Branch and Bound can be applied to this selection problem. CLO2 3
- (b) Let you have four jobs with the following penalties, deadlines and required times respectively:
 $\{p_1, p_2, p_3, p_4\} = \{5, 10, 6, 3\}$,
 $\{d_1, d_2, d_3, d_4\} = \{1, 3, 2, 1\}$ and
 $\{t_1, t_2, t_3, t_4\} = \{1, 2, 1, 1\}$
Now, generate the state space tree to sequence the jobs with deadlines using branch and bound approach. CLO3 5
7. (a) Why does the load factor of an open addressing always lie between 0 and 1? CLO2 2
- (b) In a banking system where hashing is used for storing account numbers, explain the importance of choosing a good hash function in this scenario. CLO2 3
- (c) A hash table has a size of 11 (index 0 to 10). The hash functions are defined as:
 $h_1(\text{key}) = \text{key} \bmod 11$
 $h_2(\text{key}) = 7 - (\text{key} \bmod 7)$
Using **double hashing**, insert the following keys in order: 58, 14, 91, 25, 72 CLO3 3



Faculty of Science, Engineering and Technology Department of Computer Science and Engineering

CSE 227: Data Communication

Batch: 26th

Date: 08/03/2026

Final Examination

Semester: Fall 2025

Time: 2 Hours

Total Marks : 40

Answer any five questions:

5 × 8 = 40

CLOs Marks

- | | | |
|---|------|---|
| ✗ (a) Explain Quantization in Pulse Code Modulation (PCM). | CLO3 | 4 |
| (b) State the differences among three techniques in serial transmission. | CLO1 | 4 |
| 2. (a) Briefly explain different types of digital-to-analog conversion. | CLO5 | 4 |
| (b) Define constellation diagram and explain its role in analog transmission. | CLO5 | 4 |
| 3. (a) List all categories of multiplexing. Define FDM. | CLO5 | 3 |
| (b) Illustrate three strategies of time division multiplexing. | CLO5 | 5 |
| 4. (a) Describe spread spectrum and mention its goal. | CLO5 | 4 |
| (b) Draw and explain digital signal service (Digital hierarchy). | CLO4 | 4 |
| 5. (a) How does sky propagation differ from line-of-sight propagation? Explain. | CLO1 | 4 |
| (b) Write the differences between omnidirectional waves and unidirectional waves. | CLO | 4 |
| 6. (a) Make a table for the distinct forms of UTP cables. | CLO2 | 4 |
| (b) Write the advantages and disadvantages of fiber optic cable. | CLO2 | 4 |
| ✗ (a) Explain packet switching network. | CLO3 | 4 |
| (b) Describe three phases of circuit-switched network. | CLO3 | 4 |